

Vinh Nguyen

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Melbourne, VIC | 0455 678 195 | Full Working Rights | Open to Relocate

SUMMARY

ML/AI Enthusiast with an M.S. in Artificial Intelligence from UT Austin, specializing in LLM, RAG and Agentic systems, and Deep Learning. Experienced in building end-to-end ML pipelines with MLOps tooling (DVC, MLflow, Docker), backed by a strong foundation in data analytics and operations across global organizations including adidas and New Balance, as well as start-up and family business settings.

EDUCATION

University of Texas at Austin <i>Master of Science in Artificial Intelligence</i>	Austin, TX Aug. 2024 – May 2026
• GPA: 3.93/4.0 • $\Phi K \Phi$ Honor Society	
Centennial College <i>Advanced Diploma in AI Software Eng Tech</i>	Toronto, ON Aug. 2022 – Aug. 2024
• GPA: 4.43/4.5 • CCSAI Scholarships	
Georgia Institute of Technology <i>Master of Science in Computer Science</i>	Atlanta, GA Aug. 2025 – Exp. 2028
• CGPA: 4.0/4.0	
Foreign Trade University <i>Bachelor in Economics</i>	Ho Chi Minh, VN

EXPERIENCE

Machine Learning Student Researcher <i>Centennial College</i>	Jul. 2024 – Aug. 2024 Toronto, ON
Led a 7-week exploratory ML research project on the newly released Open Oximeter Repository dataset; utilized SQL and data wrangling techniques to analyze data quality, validate availability with stakeholders, and assess opportunities for future SpO_2 estimation model development. [Project Report]	
Data Analyst (Intern) <i>New Balance Canada</i>	Jan. 2023 – Aug. 2024 Mississauga, ON
Built automated reporting tools using Excel VBA, PowerQuery, and PowerPivot, converting multi-hour manual processes into sub-10-minute automated pipelines for Sales, Accounting and Planning; modified US regional dashboards in Spotfire to enable cross-border access to business reports for the Canadian teams.	
Process and Technology Analyst <i>New Balance Vietnam</i>	Apr. 2021 – Aug. 2022 Ho Chi Minh, VN
Collaborated with global product and costing teams to analyze SOPs, optimize production workflows, and develop calculation tools, and cleared a backlog of over 300 pending issues while improving operational visibility and earning the New Balance Associate Appreciation Award.	
Strategic Project Manager <i>HEAD Phat Thinh (Honda Exclusive Authorized Dealers - Family business)</i>	Sep. 2019 – Sep. 2020 Ho Chi Minh, VN
Led a small team to launch a centralized business management system, transitioning operations from manual Excel-based tracking to a unified platform across 6 showrooms while expanding e-commerce channels, standardizing workflows, and optimizing staffing allocation based on business needs.	
Operations Manager <i>Speedia Footwear (Start-up)</i>	May. 2018 – Aug. 2019 Ho Chi Minh, VN
Built and managed a cross-functional sourcing and operations team spanning product development, design coordination, costing, material planning, molding, factory follow-up, and supplier communication for an early-stage footwear start-up from concept to production preparation.	
Costing Specialist <i>adidas Sourcing Ltd. - Liaison Office</i>	Feb. 2012 – May. 2018 Ho Chi Minh, VN
Managed product costing and enterprise costing systems to support cost optimization, production milestone tracking, process improvement, and operational compliance audits while serving as a key user and trainer for costing and product lifecycle management platforms including PLM, SATRA, ShoesCost, and ProCost.	

PROJECTS

- Light ReAct Agent with MCP Tavily & Local Tools** | *Python, LiteLLM, MCP, Tavily, GAIA* • [Code](#)
- Designed and implemented a ReAct agent framework from scratch with tool abstraction, MCP integration, event tracking, and local workspace management, enabling multi-step reasoning and tool-augmented problem solving.
 - Benchmarked against a pure LLM baseline (both use gpt-5-mini) on 30 GAIA questions (10 per each difficulty level); the agent improved accuracy by $3.75\times$ (13.33% $\rightarrow$ 50.00%) and solvable rate by $3.57\times$ (23.33% $\rightarrow$ 83.33%), at the cost of $1.9\times$ longer runtime (5m 57s $\rightarrow$ 11m 26s) and $8.6\times$ higher spend (\$0.09 $\rightarrow$ \$0.77).
- Myth-GraphRAG: Knowledge Graph for Literary QA** | *Python, ChromaDB, Neo4j, Cypher, OpenAI* • [Code](#)
- Built a GraphRAG system for The Odyssey by constructing a knowledge graph of entities and relationships, with two retrieval strategies: Global Search (community summaries) and Local Search (entity-based traversal).
 - Benchmarked on 20 questions using relevance, completeness, and faithfulness metrics; GraphRAG Local Search achieved the strongest overall performance, winning **75%** of evaluations and outperforming BaseRAG Vector Search (50%) and GraphRAG Global Search (25%).
- Multi-Document RAG Assistant** | *Python, LangChain, ChromaDB, HF, Streamlit* [Demo](#) • [Code](#)
- Developed a RAG system capable of ingesting PDF, DOCX, and TXT files; implemented text chunking, embedding generation, and semantic vector search using ChromaDB to enable precise, context-aware conversational QA.
- LLM Tool-Calling Weather Assistant** | *Python, Groq, Geopy, Open-Meteo, Streamlit* [Demo](#) • [Code](#)
- Engineered a deterministic tool-calling workflow utilizing Groq API to extract structured parameters from natural language prompts, seamlessly orchestrating Geopy and Open-Meteo APIs for real-time weather data retrieval.
- Grade School Math Reasoning App** | *HF Transformers, LoRA, PEFT, DVC, Gradio* [Demo](#) • [Code](#)
- Fine-tuned small LLMs on GSM8K using LoRA-based supervised fine-tuning and evaluated instructional few-shot Chain-of-Thought (CoT) prompting, Direct Answer SFT, and CoT SFT reasoning strategies.
- Restaurant Review Sentiment Classification** | *PyTorch, HF Transformers, DVC, Gradio* [Demo](#) • [Code](#)
- Fine-tuned DistilBERT for 5-class sentiment classification on the Yelp Review Full dataset, implementing context window filtering to optimize sequence length; engineered an end-to-end training pipeline with DVC.
- Quantization and LoRA Techniques** | *Python, PyTorch, LoRA, Quantization* • [Code](#)
- Developed from-scratch implementations of FP16/BF16 half-precision training, 4-bit quantization, and LoRA adapters for parameter-efficient LLM fine-tuning.
- Scene Text Recognition Pipeline** | *PyTorch, YOLOv11, Ultralytics, OpenCV, DVC* • [Code](#)
- Architected a two-stage Scene Text Recognition pipeline deploying YOLOv11 for text detection and a custom CRNN with CTC loss for sequence recognition, achieving a 98% word-level accuracy on the ICDAR 2013 dataset.
- Brain Tumor Segmentation** | *PyTorch, UNET, ResNet, Segmentation, IoU* • [Code](#)
- Built a UNET model with a ResNet-34 encoder trained on multi-modal MRI scans to automatically segment brain tumors, achieving an IoU score of 0.65 on the validation set.
- Self-Supervised Contrastive Learning** | *PyTorch, ResNet, Contrastive Learning* • [Code](#)
- Built a ResNet-18-based contrastive learning model for chest X-ray pneumonia classification, achieving 81.25% accuracy with limited labeled data.
- MLOps Pipeline System** | *Python, PyTorch, DVC, Mlflow, Poetry, Docker* • [Code](#)
- Built an end-to-end reproducible ML pipeline using Hydra, DVC, MLflow, Poetry, and Docker for experiment tracking, configuration management, and workflow orchestration.
- MLflow Tracking Server on AWS** | *Python, Mlflow, AWS EC2, Amazon S3, Scikit-learn, Pandas* • [Code](#)
- Built an MLflow Tracking Server using AWS EC2, and S3 to enable remote experiment tracking, and artifact storage from local training workflows.

TECHNICAL SKILLS

GenAI & LLMs: HF Transformers, Fine-tuning, PEFT (LoRA), Quantization, GRPO, RLVR, RAG, GraphRAG, Agentic AI, LangChain, Neo4j

Machine Learning: Python, PyTorch, Scikit-learn, YOLO, NumPy, Pandas, Matplotlib

Data Engineering: SQL, MongoDB, ChromaDB, PowerBI, PowerQuery, PowerPivot, Tableau, Spotfire

MLOps & Cloud: Docker, Git, GitHub Actions, CI/CD, MLflow, DVC, Hydra, FastAPI, AWS, Streamlit, Gradio